

$$\begin{array}{l} A: \neg p \wedge q \\ B: r \vee p \\ C: \neg r \rightarrow s \\ D: s \rightarrow t \\ \hline \therefore t \end{array}$$

$$[(\neg p \wedge q) \wedge (r \vee p) \wedge (\neg r \rightarrow s) \wedge (s \rightarrow t)] \rightarrow t$$

p/q addition.
 $\therefore p \vee q$

p/q simplification
 $\therefore p \vee q \quad \therefore q$

p
p $\rightarrow$ q
 $\therefore q$

$$[p \wedge (p \rightarrow q)] \rightarrow q$$

$\neg q$
p $\rightarrow$ q
 $\therefore \neg p$

$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$ Modus Tollens

p $\rightarrow$ q
q $\rightarrow$ r
 $\therefore p \rightarrow r$

hypothetical syllogism

p $\vee$ q
 $\neg p$
q

Disjunctive syllogism

p
q
 $\therefore p \wedge q$

conjunction

p $\vee$ q
 $\neg p \wedge r$
 $\therefore q \vee r$

resolution

Solution for (1) (using rules of inference)

- | Step | Reason |
|---------------------------|-------------------------|
| 1. $\neg p \wedge q$ | Premise A |
| 2. $\neg p$ | Simplification |
| 3. $r \vee p$ | Premise B |
| 4. $\neg r$ | Modus Tollens (2 and 3) |
| 5. $\neg r \rightarrow s$ | Premise C |
| 6. s | |
| 7. $s \rightarrow t$ | Premise D |
| 8. t | |

Rules of Inference
Methods of proving theorem

- Direct proof

- Proof by contraposition

- Proof by contradiction

- Proof by equivalence

- Proof by exhaustion

- Principle of mathematical induction

- Principle of complete induction

A: $\neg p \wedge q$

B: $q \rightarrow p$

C: $\neg p \rightarrow q$

D: $p \rightarrow \neg q$

E: $\therefore t$ (conclusion)

$\forall x P(x)$

$\therefore P(a)$ is true for some a in domain of $P(x)$

$P(c)$ is true for some arbitrary c

$\therefore \forall x P(x)$

$D(x)$: x is a student of CS-205

$E(x)$: x is a student enrolled for CS.

$\forall x (D(x) \rightarrow E(x))$

Every student who has taken CS-205 has enrolled for CS.

Direct proof

p : n is odd, $n \in \mathbb{Z}^+$

$n = 2k+1, k \in \mathbb{Z}^+$

q : n^2 is odd

$n^2 = (2k+1)^2 = 4k^2 + 4k + 1$
 $= 2(2k^2 + 2k) + 1$

odd
 $2k$

for $n \in \mathbb{Z}^+$, if $3n+2$ is odd, then n is odd.

$3n+2 = 2k+1$

$3n = 2k-1$

$n = \frac{1}{3}(2k-1)$

Proof by contraposition

$p \rightarrow q \equiv \neg q \rightarrow \neg p$

$\neg q$: $3n+2$ is even

$\neg p$: n is even

$\neg D(x)$: n is even

$n = 2k, k \in \mathbb{Z}^+$

$\therefore \neg D(x) \rightarrow \neg p$

$\downarrow$

$\neg p \rightarrow \neg D(x)$

$\neg p$: $3n+2$ is even

$3(2k)+2 = 2(3k+1) \rightarrow$ even



if $m = ab$, $a \in \mathbb{Z}^+$, $b \in \mathbb{Z}^+$
then $a = \sqrt{m}$ or $b = \sqrt{m}$

$$\sim (a = \sqrt{m} \text{ or } b = \sqrt{m})$$

$$\sim (a = \sqrt{m}) \cap \sim (b = \sqrt{m})$$

$$a \neq \sqrt{m} \cap b \neq \sqrt{m}$$

$\therefore a \neq \sqrt{m}$ and $b \neq \sqrt{m}$
Then product $ab \neq m$

$$\therefore \sim \neg \rightarrow \sim p$$

$$\Rightarrow p \rightarrow q$$

Proof: $\sqrt{2}$ is irrational

$$\nexists p/q \mid \frac{p}{q} = \sqrt{2} \vee p, q \in \mathbb{Z}^+ \quad q \neq 0$$

$$p \rightarrow q \text{ and } (p \wedge (\neg q)) \rightarrow F$$

$$\begin{array}{r} p \rightarrow q \\ \sim q \\ \hline \therefore \sim p \end{array}$$

$\Rightarrow \sqrt{2}$ is not rational

Assume $\sqrt{2}$ is rational

$$\sqrt{2} = \frac{p}{q} \quad p \text{ and } q \text{ are lowest factors}$$

$$\frac{p^2}{q^2} = 2 \quad \frac{p^2}{q^2} = \frac{4k^2}{2} = 2k^2$$

$$p^2 = 2q^2$$

$$p^2 \text{ is even} \Rightarrow q^2 \text{ is even}$$

$$\Rightarrow p \text{ is even}$$

p and q have a common factor 2.
 $\therefore \sqrt{2}$ is not rational.

$\forall n \in \mathbb{Z}$; n is odd $\Rightarrow n^2$ is odd.

p : n is odd

Σ : n^2 is odd

$$(p \rightarrow q) \wedge (q \rightarrow p) \equiv p \leftrightarrow q$$

$p \rightarrow q$

$$n = 2k+1$$

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 \Rightarrow \text{odd}$$

$$\Sigma \rightarrow p \quad \text{---} \quad n = 2k$$

$$n^2 = 4k^2 = 2 \cdot 2k^2$$

$$p \rightarrow q \quad \text{---} \quad q \rightarrow p$$

$\therefore (p \rightarrow q) \wedge (q \rightarrow p)$ is true.

Proving cases:

1. Every $x \in \mathbb{Z}^+$ can be represented as sum of squares of 2 integers.

$x=3$ (counter example that disproves the above)

2. Every $n \in \mathbb{Z}$, prove that $n^2 \geq n$ p. remine)

Case 1. p_1 : n is negative

Case 2. p_2 : $n=0$.

Case 3. p_3 : n is positive

$$|x| \geq |y| \Rightarrow |x| \cdot |y|$$

$$p_1: x \geq 0; y \geq 0$$

$$p_2: x \geq 0; y \leq 0$$

$$p_3: x \leq 0; y \geq 0$$

$$p_4: x \leq 0; y \leq 0$$

$$x^2 + 3y^2 = 8, x \in \mathbb{Z}$$

$$x^2 = 8 - 3y^2$$

$$x = \{-2, -1, 0, 1, 2\}$$

$$y = \{-1, 0, 1\}$$

Check for greatest values of x and y

$$(n+1)^2 \geq n \quad \text{for } n \in \mathbb{N}$$

$$2^2 \geq 1 \quad \checkmark$$

$$9 \geq 2 \quad \checkmark$$

$$16 \geq 3 \quad \checkmark$$

$$25 \geq 4 \quad \checkmark$$

(proof by exhaustion, where possible values can be listed)

contradiction $\therefore$ 22 days are chosen
 $p \rightarrow q \equiv p \vee \neg q$ $\therefore$ At least four of 22 days that are chosen fall
 on same day of the week
 $\neg q$: At most 3 days ~~are~~ chosen fall on same
 day of the week.

Rules of inference	Step	reason
$A: (p \wedge q) \vee r$	1. $(\neg p \wedge q) \vee r$	remove A -
$B: x \rightarrow y$	2. $(p \vee x) \wedge (q \vee x)$	expansion of
$\underline{p \vee x}$	3. $x \vee p$	simplification of (2)
	4. $x \rightarrow y$	
	5. $x \vee y$	
	6. $p \vee y$	resolution of 3 and 5

there exists
 - Prove that only 2 consecutive integers < 100
 that are perfect powers of integers

proof by enumeration

power of 2 : 1, 4, 9, 16, 25, 36, 49, 64, 81

" 3 : 1, 8, 27, 64

4 : 1, 16, 81

5 : 1, 32

6 : 1, 64

first no. to be expressed as sum of cubes
in diff. ways

$$1729 = 10^3 + 9^3$$

$$= (2^3 + 1)$$

For $P(n)$ by (x+y) $\rightarrow$ NP(y)
lower

Given $a+b = x$

1. $a^2 = ab$

2. $a-b^2 = ab-b$

3. $(a-b)(a+b) = b(a-b)$ wrong

4. $a+b = 5$

5. $ab = b$

6. $2 = 1$

(denying the hypothesis)

~~1729~~

Hanks algorithm $\rightarrow$ (3n+1) transformation

~~For collatz~~

$$T(x) = \begin{cases} x/2 & \text{if } x = \text{even} \\ 3x+1 & \text{if } x = \text{odd} \end{cases}$$

$T(26) = 13$

$T(13) \rightarrow 40$

$T(40) = 20$

$T(20) \rightarrow 10$

$T(10) \rightarrow 5$

$T(5) \rightarrow 16$

$T(16) = 8$

$T(8) = 4$

$T(4) = 2$

$T(2) = 1$

Hanks algorithm reduces every number to 1 (irrespective of value of x)

PM1 example

$\forall x (x \in \mathbb{Z}^+) \rightarrow x! \leq x$

$\forall x (x \in \mathbb{Z}^{++}) \rightarrow 3x^3 - x$

given a series of x's $a_1, a_2, a_3, \dots, a_n$

by PM1
Ext. $P(x)$ is the assumption that $P(x)$ is true

* For $n \geq 1$, prove that $n! \geq 2^{n-1}$

$$P(n) : n! \geq 2^{n-1}$$

$$P(1) : 1! \geq 2^{1-1} = 2^0 = 1$$

$$P(k) : k! \geq 2^{k-1}$$

$$P(k+1) : (k+1)! \geq (k+1)k! \geq (k+1)2^{k-1}$$

$$k+1 \geq 2$$

$$\geq 2 \cdot 2^{k-1} = 2^k$$

Function $f(A)$ & $\{$ $P(n) = Cn + ADn$
 $C \in \mathbb{R}$
 $D \in \mathbb{R}$

while $(D \neq A)$ $\{$
 $C \leftarrow C + A$;
 $D \leftarrow D + 1$;
 $\}$
 return C ;

Complete induction

$$P(1) \wedge P(2) \dots \wedge P(n) \rightarrow P(n+1)$$

→ Proof by PMI
 $2^n < n!$ (for $n \geq 4$)
 $3 < n^3 - n$

ALDEAF

$$P(n) \Rightarrow C_k = D * A_k$$

SS(A)

1. $C = 0$

2. $D = 0$

3. while ($C \neq n$)

$$C \leftarrow C + 1$$

$$D \leftarrow D + 1$$

4. return C

$$P(0) \Rightarrow C = 0, D = 0$$

$$P(k) \Rightarrow C_k = D A_k$$

$$P(k+1) \Rightarrow C_{k+1} = D A_{k+1}$$

A recurrence relation is a linear homogeneous relation of order k if it can be expressed in the form -

$$a_n = \mu_1 a_{n-1} + \mu_2 a_{n-2} + \dots + \mu_k a_{n-k}$$

where μ_i 's are constants.

$$C_n = -2 C_{n-1} \text{ [order} \rightarrow 1]$$

$\mu_i \in \mathbb{R} \text{ } i=1, 2, \dots, k$

$$f_n = f_{n-2} + 4 f_{n-1} \text{ [order} \rightarrow 2]$$

$$a_n = a_{n-1} + 3 \text{ (not a linear homogeneous relation)}$$

$$g_n = g_{n-1}^2 + g_{n-2}$$

$$x^n = r_1 x^{n-1} + r_2 x^{n-2} + \dots + r_{n-1} x^1 + r_n x^0$$

characteristic eqⁿ of $\begin{pmatrix} r_{n-2} & r_{n-1} \\ r_{n-1} & r_n \end{pmatrix}$

if s_1 and s_2 are roots of characteristic eqⁿ,
distinct

$$a_n = u s_1^n + v s_2^n$$

u, v are constants

u & v depend on initial conditions

if $s_1 = s_2$ $a_n = u s_1^n + n v s_1^{n-1}$

$$C_n = 3C_{n-1} - 2C_{n-2}$$

$$x^2 = 3x - 2$$

characteristic eqⁿ

$$x^2 - 3x + 2 = 0$$

$$x^2 - 2x - x + 2 = 0$$

$$x(x-2) - (x-2) = 0$$

$$x^2 + 2$$

$$s_1 = 2, s_2 = 2$$

$$a_n = u \cdot 1^n + v \cdot 2^n = u + 2v$$

$$5 = u + 2v$$

$$u + 4v = 3$$

$$\frac{u + 4v = 3}{u + 2v = 5}$$

$$C_2 = u \cdot 1^2 + v \cdot 2^2 = 5 = u + 4v$$

$$v = -1$$

$$a_n = u s_1^n + v s_2^n = 7(1)^n + (-1)2^n = 7 - 2^n$$

$$\rightarrow d_n = 2d_{n-1} - d_{n-2}$$

$$x^2 = 2x - 1$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$s_1, s_2 = 1$$